Cristian Mejia

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Education

Stanford University Stanford, CA

B.S. in Mechanical Engineering, Intended Minor in Aeronautics & Astronautics; GPA: 3.66 *Expected June 2029*

- **Awards:** QuestBridge Scholar (Full Scholarship Recipient), College Board National Hispanic Recognition Program, College Board National First-Generation Recognition Program
- **Relevant Coursework:** Solid Mechanics, Mechanics (Physics), Vector Calculus, ODEs, Linear Algebra & PDEs, Programming Methodology (Python)
- **Upcoming Coursework:** Engineering Thermodynamics, Introductory Fluids Engineering, Mechanics of Materials, Dynamics, Foundations of Product Realization, Product Realization: Design and Making, Electricity and Magnetism, Programming Abstractions, Introduction to Computer Graphics and Imaging

Experience

Mechanical Engineer, Dynamics Team Sept. 2026 – Present

Stanford Solar Car Project *Stanford, CA*

- Design and build parts for Sunstruck, the team's car for the 2027 World Solar Challenge in Australia, using CAD, 3D printing, and CNC machining
- Designed a water-resistant motor controller enclosure in Fusion 360 for 3D printing in ASA, sealed with a closed-cell EPDM foam gasket and mounted to the carbon fiber chassis with M6 bolts into threaded inserts
- Projected the controller's connectors onto the enclosure wall and offset each cutout 0.5 mm for clearance
- Designing a drill jig to cross-drill roll pin holes through tubes with bonded aluminum inserts, clocking the holes at each end 90° apart and spacing them precisely from the tube ends to avoid damaging the inserts

Mathematics and Physics Tutor 2026 – Present

Shires Tutoring *Stanford, CA*

- Tutor high school students in math through Calculus BC and introductory physics using custom lesson plans

Licensing Intern Aug. 2026

JAKKS Pacific *Santa Monica, CA*

- Researched AI toy trends and delivered an informational presentation on AI and its integration into toys
- Analyzed market data and presented business cases and licensing strategies to the marketing team

Team Member Oct. 2025 – Dec. 2025

Stanford Student Space Initiative *Stanford, CA*

- Soldered and reworked electronics and contributed to mission documentation for SAMWISE, a 2U CubeSat

Projects

Automated Task Management System | *ESP32, Fusion 360, LLM API* Los Angeles, CA

- Built an ESP32-based task manager, wiring the circuit and integrating a microphone for real-time audio capture
- Designed a hinged enclosure in Fusion 360 that mounts the display and conceals all internal electronics and wiring
- Programmed a speech-to-text and LLM API pipeline that extracts tasks and displays notifications, resolving hardware-software integration issues for low-latency audio streaming

Wind Turbine Design | *CEE 34N: Wind Energy Explained* Stanford, CA

- Led airfoil blade fabrication and 3D printing on a 3-person team, including the generator's PLA shaft
- Helped redesign the rotor after the airfoil blades couldn't self-start; the final high-solidity drag rotor lifted ∼36 steel rods (∼0.36 kg) ∼0.9 m in ∼10 s
- Co-authored a report estimating ∼0.32 W output and a ∼3% power coefficient (vs. the 59% Betz limit) from a measured ∼50 rpm rotor speed

Technical Skills

Programming: Python, MATLAB
CAD & Fabrication: Fusion 360, 3D Printing, CNC Machining
Electronics: Soldering & Rework, Circuit Wiring, ESP32 Microcontroller